Methodology

Recruitment Funnel Performance Review - Multi-Property Hospitality Recruitment (Jan–Jun 2026)

Approach

- Analyzed a 700-candidate recruitment funnel dataset spanning six sourcing channels, four recruiters, five departments, five properties, and three locations, using Python (pandas) for data cleaning and exploratory analysis and Power BI for interactive dashboarding.
- Structured the analysis as a progressive drill-down, overall funnel to dimension-level performance (source, recruiter, department, property, location) to interaction effects (recruiter × source, recruiter × hiring manager) to hiring velocity to diversity trends to correlation checks.

Assumptions

- Each Days to X metric (Screen, Interview, Offer, Join) was treated as valid only for candidates who actually reached that corresponding stage; values present for candidates who did not reach the stage were treated as invalid and excluded from averages.
- Property and Location were analyzed as independent dimensions rather than a combined "property-in-its-city" view, given inconsistent mapping between the two fields in the source data.
- Findings drawn from small-sample cuts (e.g., Campus and Social Media sources at 33 interviews each; Offer Decline Reason at 16 total entries) are reported as directional rather than conclusive due to limited volume.
- Requisition Ageing Days was treated as a requisition-level sourcing-speed metric, conceptually distinct from candidate-level funnel velocity.
- Requisition Ageing Days showed negligible correlation with candidate-level velocity metrics and was therefore excluded from the dashboard as non-actionable, though this is retained in the notebook.
- Offer Decline captured candidates who explicitly declined an offer consistent with the Offer Decline Reason field (16 entries, matching the explicit-decline count exactly).
- Drop-off reflects candidates who accepted but did not ultimately join a distinct behaviour not to be combined with the active decline decision.

Data Quality Observations

- Requisition Close Date was 100% null across all 700 records and was dropped from the analysis.
- All four Days to X columns contained non-null values even for candidates who never reached the corresponding stage. Affected values were nulled before calculating averages.
- Each Property appeared across all three Location values in similar proportions.
- No duplicate candidate records and no missing values were found on any core categorical or funnel-stage field.

Analytical Techniques Applied

- Conditional aggregation and stage-to-stage conversion analysis to quantify funnel drop-off at each transition.
- Progressive segmentation (source to recruiter to hiring manager) to isolate interaction effects rather than relying on single-dimension averages.
- Comparative analysis of conversion rate, offer-decline rate, and average processing time across Source, Recruiter, Department, Property, Location, Diversity Category, Gender, and Age Band.
- Pearson correlation analysis to test relationships between requisition ageing and candidate-level velocity metrics, and among stage-to-stage processing times.
- Interactive dashboarding (Power BI) with cross-filtered slicers, a funnel visualization, and combo-chart views to surface interaction effects and quality-versus-speed changes visually.